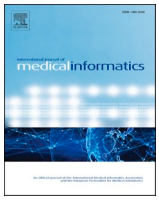
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Impact of a non-interruptive electronic health record-based clinical decision support tool on HEART score documentation compliance in the emergency department

Eric Boccio [a,b,*], Randy Katz [c], Michael Weiss [d]

[a] Department of Emergency Medicine, Mount Sinai Medical Center of Florida, Miami Beach, FL 33140, USA
[b] Department of Emergency Medicine, New York Medical College – School of Medicine, Valhalla, NY 10595, USA
[c] Department of Emergency Medicine, Memorial Healthcare System, Hollywood, FL 33021, USA
[d] Department of Pediatric Emergency Medicine, Joe DiMaggio Children's Hospital, Hollywood, FL 33021, USA

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ABSTRACT

Objective: To evaluate whether implementation of a non-interruptive, workflow-integrated electronic health record (EHR) clinical decision support (CDS) tool was associated with a sustained increase in HEART score documentation rates relative to prior manual, provider-driven documentation practices.

Methods: A quasi-experimental study employing an interrupted time series (ITS) design was performed across six ED entities within a single health system in the southeastern United States. Eligible patients were aged ≥21 years, carried an ICD-10 code for undifferentiated cardiac chest pain (R07.1, R07.2, R07.9, R07.81, R07.82, R07.89), and had an electrocardiogram performed and a troponin resulted during their ED length of stay. A non-interruptive, note-embedded CDS tool was announced in October 2023 and implemented system-wide in November 2023. The pre-implementation period spanned January 2022 through September 2023 (21 months); the post-implementation period spanned January 2024 through March 2026 (27 months); October through December 2023 served as a washout window. Monthly documentation compliance rates were compared using the Wilcoxon rank-sum test. Additionally, an ITS analysis using segmented regression was performed to estimate the immediate level change and trend slope change at the point of program implementation.

Results: A total of 81,895 patient encounters were eligible for inclusion during the study period. The median pre- and post-implementation monthly compliances were 29.0% (95% CI: 28–30%) and 82% (95% CI: 81–83%), respectively, and the difference between cohorts was statistically significant ($Z = -5.92$, $p < 0.0001$). The pre-implementation monthly compliance trend was modest and positive ($\beta = 0.191$ percentage points/month, 95% CI: 0.042–0.340 percentage points/month, $p = 0.012$); there was no statistically significant change in slope following implementation ($\beta = -0.114$ percentage points/month, 95% CI: −0.303 to 0.075 percentage points/month, $p = 0.237$). A robust level change at the intervention point was found to be statistically significant ($p < 0.0001$).

Conclusion: A non-interruptive, workflow-embedded EHR CDS tool was associated with a rapid and durable 53-percentage-point increase in HEART score documentation compliance across a multi-site ED system. Non-interruptive tools that embed decision support within existing clinical documentation workflows represent a sustainable informatics approach to improving provider adherence to evidence-based cardiac risk stratification.

1. Introduction

Chest pain is among the most frequent reasons for emergency department (ED) visits, accounting for approximately 7 million annual presentations in the United States [1]. Despite this high volume, only a minority of these encounters ultimately results in a diagnosis of acute coronary syndrome (ACS); however, missed occlusive myocardial infarction remains among the most consequential diagnostic errors in emergency medicine, contributing to preventable morbidity, mortality, and significant medicolegal liability [2]. Balancing timely identification

* Corresponding author at: Department of Emergency Medicine, Mount Sinai Medical Center of Florida, 4300 Alton Road, Miami Beach, FL 33140, USA.
E-mail address: bocciomedicalservices@gmail.com (E. Boccio).

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of high-risk patients with the safe discharge of low-risk patients is a central and persistent challenge for emergency physicians in a time-pressured, high-acuity environment.

The HEART (History, EKG, Age, Risk factors, Troponin) score is a validated clinical decision support (CDS) tool designed to estimate the short-term risk of major adverse cardiac events (MACE) in patients with undifferentiated chest pain [3]. Stratification into low- (0–3), intermediate- (4–6), and high-risk (≥7) categories guides diagnostic testing, therapeutic interventions, resource utilization, and disposition decisions. The HEART score has been endorsed by both the American College of Cardiology (ACC) and the American College of Emergency Physicians (ACEP) as a preferred risk stratification tool for patients with suspected ACS when using conventional or select high-sensitivity troponin assays [5,6]. Despite its strong evidence base and guideline support, consistent utilization and documentation of the HEART score in real-world ED clinical practice remains suboptimal, with studies demonstrating wide variability in provider uptake across sites and practice settings [7].

Electronic health record (EHR)-embedded CDS tools have emerged as a strategy to improve provider adherence to evidence-based practices across clinical domains [8]. In the ED, CDS interventions have targeted medication dosing, diagnostic imaging appropriateness, sepsis identification and management, and guideline-concordant documentation [9]. However, many traditional implementations rely on interruptive best practice advisory (BPA) alerts, which require active provider acknowledgment, generate significant alert fatigue, and are bypassed by clinicians in up to 90–95% of encounters [10,11]. Non-interruptive, workflow-embedded CDS tools deliver decision support within the natural flow of clinical documentation without requiring a forced interaction or hard stop and may represent a more sustainable mechanism for influencing provider behavior at scale.

Structured documentation of the HEART score in the EHR confers benefits that extend beyond the individual clinical encounter: it enables quality monitoring and cross-site benchmarking, supports medicolegal documentation of medical decision making, facilitates system-level accountability, and creates infrastructure for longitudinal outcomes research on chest pain pathway effectiveness [12]. The objective of this study was to evaluate the effect of implementing a non-interruptive, note-embedded EHR CDS tool on provider HEART score documentation compliance across a multi-site ED system over a 4-year period. We hypothesized that deployment of this tool would be associated with a significant, rapid, and sustained improvement in HEART score documentation compliance.

2. Related work

The HEART score was introduced by Six et al. in 2008 and subsequently validated by Backus et al. in a prospective multicenter cohort and demonstrated a negative predictive value (NPV) of 98.3% and c-statistic of 0.83 for 6-week MACE prediction [3,4,13]. A systematic review and meta-analysis by Fernando et al. confirmed 95.9% sensitivity at a HEART score ≥4 and 95.0% specificity at ≥7, with the HEART score outperforming both TIMI and GRACE for ED chest pain risk stratification [14]. A meta-analysis of low-risk patients demonstrated an aggregate MACE rate of approximately 1.7–2.0% at 30 days for scores 0–3, supporting the safety of early discharge in this population [15].

Building on the basic score, Mahler et al. developed the HEART Pathway, which combines HEART score ≤3, a nonischemic EKG, and serial troponin measurements to identify low-risk patients eligible for early discharge, achieving an NPV of 99.6% for 30-day MACE [16]. A stepped-wedge, cluster randomized trial by Poldervaart et al. demonstrated that HEART score implementation was associated with significant changes in early hospital discharge of low-risk patients without increasing adverse outcomes [17]. The 2022 ACC Expert Consensus Decision Pathway and the 2021 American Heart Association/ACC Chest Pain Guideline both endorse structured risk stratification pathways for ED chest pain, with the HEART score cited among the list of preferred instruments [5,18].

Despite strong guideline support, implementation studies have consistently highlighted gaps in real-world provider utilization. Inter-rater reliability studies have demonstrated good overall agreement, with greatest variability in subjective components (history and EKG interpretation), further highlighting the value of structured, EHR-embedded data capture tools to standardize scoring [19].

A systematic review by Kwan et al. examining EHR-integrated CDS in the ED identified 42 eligible studies, of which 83% reported positive effects on process outcomes including improved guideline adherence, appropriate ordering practices, and documentation completeness [9]. The review noted that many interventions relied on uncontrolled pre-post designs and highlighted the feasibility of more rigorous evaluations using interrupted time series analyses. A comprehensive systematic review of information systems in ED decision-making further underscored that passive, analytically-timed systems as opposed to active interruptive alerts are particularly well-suited to the high-acuity, time-restricted ED environment [20].

Alert fatigue is the primary barrier to CDS effectiveness in clinical environments. Ancker et al. demonstrated that repeated, high-volume alerts significantly diminish provider response rates [10]. Alexiuk et al. identified four core barriers to EHR-CDS integration: alert fatigue, limited interoperability, poor workflow alignment, and inadequate provider trust in algorithmic outputs [11]. Similarly, Kawamoto et al. identified features predictive of CDS success: automatic provision of support within clinician workflow, recommendations at the time and location of decision-making, actionable advice, and computer-based delivery [21]. Non-interruptive, workflow-embedded tools that deliver support within existing documentation templates align with all of these principles. In the specific domain of cardiac risk stratification, the EvidencePoint platform demonstrated a non-interruptive CDS tool adoption rate of 77.8% across multiple sites, reinforcing that workflow-embedded design can achieve high utilization without interruptive alerts [21]. Mark et al. demonstrated the feasibility of automated retrospective HEART score calculation from EHR data elements [12], while Wright et al. found that EHR-integrated structured documentation tools improved completeness and consistency of documentation across provider groups [23]. To our knowledge, no prior published study has evaluated the impact of a non-interruptive, note-embedded CDS tool on HEART score documentation using an interrupted time series design spanning multiple years of longitudinal data.

3. Methods

3.1. Study design and setting

A quasi-experimental study was conducted employing a pre-post interrupted time series (ITS) design involving the retrospective review of pre-existing administrative data. The study was conducted across six ED entities comprising the Memorial Healthcare System (MHS), a large, non-profit health system located in Broward County, Florida. Monthly aggregate compliance data spanned January 2022 through March 2026 (52 months total), with October through December 2023 excluded as a washout window coinciding with tool announcement and provider education prior to implementation. This study was reviewed and approved by the Memorial Healthcare System Institutional Review Board (#MHS.2024.039).

3.2. Inclusion and exclusion criteria

Patients presenting to any of the six MHS ED entities were screened for eligibility at the encounter level. Inclusion criteria were: (1) age ≥21 years; (2) an ICD-10 code consistent with undifferentiated cardiac chest pain (R07.1, R07.2, R07.9, R07.81, R07.82, or R07.89); (3) electrocardiogram (EKG) completed during ED length of stay; and (4) troponin

resulted during ED length of stay. Both EKG completion and troponin testing were required as inclusion criteria because they are prerequisite data elements for HEART score calculation and indicate provider consideration of cardiac pathology. Patients were excluded if the ED disposition was against medical advice (AMA) or left before treatment completed (LBTC). Encounters with diagnoses indicating an established or confirmed cardiac etiology (ST-segment elevation myocardial infarction, non-ST-segment elevation myocardial infarction, unstable angina) or a clearly non-cardiac alternative diagnosis (pulmonary embolism, pneumonia, hemothorax, pneumothorax, empyema, trauma, gallstones, pancreatitis) were similarly excluded, as HEART score calculation would not constitute an appropriate clinical indication in these encounters.

3.3. Intervention

Prior to program implementation, HEART score calculation and documentation were expected to be completed manually by ED providers for all eligible encounters, and no inline or automated workflow support existed. The CDS tool was formally announced to all providers across all six MHS ED sites in October 2023 and deployed system-wide in November 2023. Unlike conventional interruptive BPA alerts that generate pop-up windows requiring acknowledgment, this tool was designed to be non-interruptive and workflow-embedded. When a patient met eligibility criteria and no HEART score had yet been filed, a hyperlink appeared within the ED Course section of the provider's existing progress note template with the prompt: 'HEART SCORE REQUIRED. CLICK HYPERLINK.' Clicking the hyperlink opened a structured flowsheet within the EHR (Epic Systems, Verona, WI), guiding the provider through discrete data entry for each of the five HEART score components (History, EKG, Age, Risk Factors, Troponin). Upon completion, the calculated total score and component-level values automatically populated back into the provider's note, replacing the prompt. The tool allowed full clinician override, maintaining that disposition and management decisions should incorporate comprehensive clinical judgment beyond the algorithmic output.

3.4. Data collection

Data were extracted from the *MHS HEART Score TeamHealth Tableau Dashboard*, a purpose-built quality improvement reporting platform developed by the MHS Informatics Technology department. The Dashboard was audited for validity prior to use, with individual parameter values reviewed to confirm alignment with the specified inclusion and exclusion criteria. All data were de-identified and aggregated at the monthly encounter level. Data elements extracted included: monthly eligible encounter counts, HEART score risk group classification (low-, moderate-, high-risk), documentation status, and ED clinical site location.

3.5. Outcome measures

The primary outcome was the presence or absence of a documented, calculated HEART score in the ED provider note, expressed as a monthly percentage compliance rate. The pre-implementation cohort encompassed January 2022 through September 2023 (21 monthly observations); the post-implementation cohort encompassed January 2024 through March 2026 (27 monthly observations).

3.6. Statistical analysis

The normality of each cohort's distribution was formally assessed using the Shapiro-Wilk test. Monthly compliance rates between cohorts were first compared using the Wilcoxon rank-sum test. An ITS analysis was conducted using a generalized estimating equations (GEE) segmented regression model with identity link function and independent correlation structure. The model included: (1) a cohort indicator (pre- versus post-) to estimate intercepts; (2) an observation number within each cohort (to estimate within-cohort monthly trend slopes); and (3) a cohort*observation interaction term (to estimate the change in slope between cohorts). Month was specified as the subject-level clustering variable. A two-sided p-value of < 0.05 was considered statistically significant. All statistical analyses were performed using SAS v9.4 (SAS Institute, Inc., Cary, NC).

3.7. Sensitivity analyses

Because the pre-implementation observation window began in January 2022, a period that may still have reflected residual pandemic-related disruption to ED operations, staffing, and case volumes predating the true 2021 pandemic peak, all primary analyses (Wilcoxon rank-sum test and GEE segmented regression) were repeated after excluding all calendar-year 2022 observations, restricting the pre-implementation cohort to January 2023 through September 2023 (9 monthly observations). The post-implementation cohort (January 2024 through March 2026) was unaffected by this exclusion.

To evaluate whether a concurrent change in ED operational volume or staffing-adjusted workload could confound the observed change in documentation compliance, monthly aggregate ED encounter volume and a weighted patients-per-hour (PPH) staffing-intensity metric were obtained for the same health system. These metrics were compared across the pre- and post-implementation windows using the segmented-regression model structure applied to the primary outcome.

4. Results

A total of 81,895 patient encounters were screened and eligible for inclusion during the study period. The Shapiro-Wilk test confirmed non-normality in both the pre- and post-implementation groups ($p = 0.027$ and $p = 0.034$, respectively). The median pre- and post-implementation monthly compliances were 29.0% (95% CI: 28–30%) and 82% (95% CI: 81–83%), respectively (Table 1). The absolute difference in median monthly compliance between cohorts was 53.0 percentage points, and this difference was statistically significant ($Z = -5.92$, $p < 0.0001$).

Fig. 1 illustrates the monthly compliance rates for eligible encounters across the full study period, stratified by cohort.

During the pre-implementation period, monthly documentation compliance demonstrated a modest but statistically significant positive trend ($\beta = 0.191$ percentage points/month, 95% CI: 0.042–0.340 percentage points/month, $p = 0.012$), reflecting a secular upward drift of approximately 0.2 percentage points per month prior to the CDS tool deployment. Immediately following CDS tool deployment, a statistically significant and clinically large level change was observed: the estimated post-implementation intercept was 79.3% (95% CI: 75.0–83.7%, $p < 0.0001$), representing the predicted compliance at the first post-implementation observation whereas the 82% median reflects the sustained central tendency of performance across all 27 post-implementation months. The post-implementation monthly slope was 0.077 percentage points/month (95% CI: −0.040 to 0.194 percentage points/month, $p = 0.197$), which was not statistically significantly different from the pre-implementation slope ($p = 0.237$). Table 2 presents the model's intercept and slope estimates while Fig. 2 displays the GEE segmented regression fit lines with 95% confidence limits for both cohorts.

Post-implementation compliance remained consistently elevated throughout the 27-month observation window. Monthly rates ranged from 76% to 86%, with 25 of 27 months (93%) achieving compliance $\geq$ 80%. The post-implementation monthly documentation compliance rate exhibited no statistically significant downward trend (slope $\beta = 0.077$ percentage points/month, $p = 0.197$), indicating stable compliance rather than progressive attenuation. Post-implementation gains cannot be attributed to a continuation of the pre-existing secular trend as the

Table 1
Descriptive statistics and Wilcoxon rank-sum comparison of monthly HEART score documentation compliance: pre- vs. post-implementation.

Cohort	n (months)	Mean (%)	SD (%)	Median (%)	Range (%)	p-value
Pre-Implementation (Jan 2022–Sep 2023)	21	29.0	2.0	29.0	24–34	<0.0001*
Post-Implementation (Jan 2024–Mar 2026)	27	82.0	2.2	82.0	76–86	—

* Wilcoxon rank-sum test; Z = −5.92; SD = standard deviation.

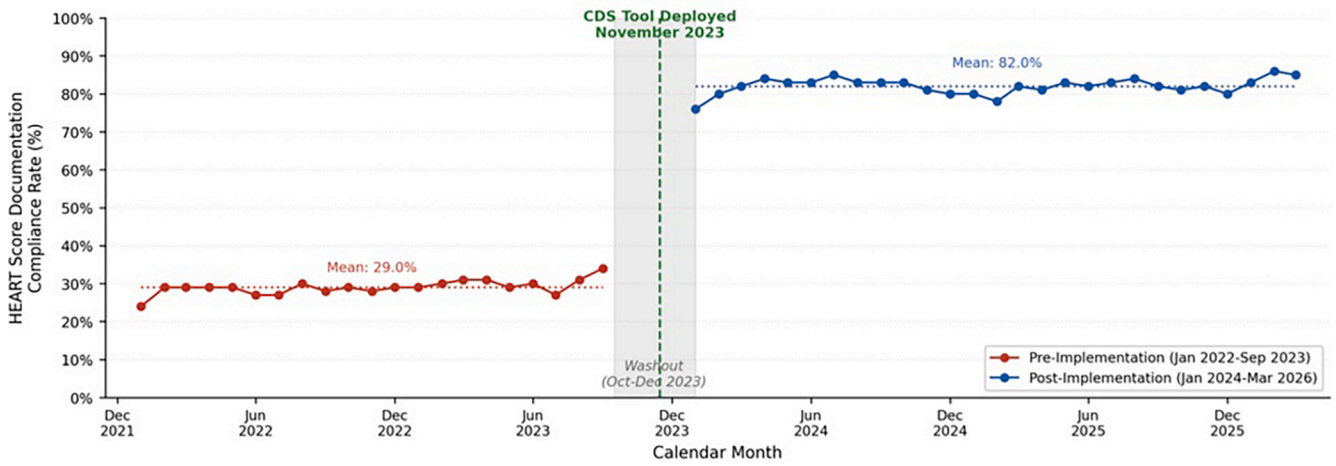


Fig. 1. Monthly HEART score documentation compliance rate (%) for eligible encounters by implementation period (January 2022–March 2026).

Table 2
Segmented regression parameter estimates — interrupted time series analysis of monthly HEART score documentation compliance.

Parameter	Estimate (%)	SE	95% CI (%)	Z	p-value
Pre-Implementation Intercept	26.95	0.87	25.24, 28.66	30.86	<0.0001
Pre-Implementation Monthly Slope	0.191	0.076	0.042, 0.340	2.52	0.012
Post-Implementation Intercept	79.34	2.24	74.96, 83.73	35.48	<0.0001
Post-Implementation Monthly Slope	0.077	0.060	−0.040, 0.194	1.29	0.197
Change in Slope at Time of Intervention	−0.114	0.097	−0.303, 0.075	−1.18	0.237

SE = empirical standard error; CI = confidence interval.

estimated secular effect extrapolated 27 months would account for an increase of approximately 5 percentage points, far below the observed 53 percentage point level change.

4.1. Sensitivity analyses

With 2022 excluded, the pre-implementation cohort was restricted to 9 monthly observations from January 2023 through September 2023 (mean 30.2%, SD 1.9%, median 30.0%, range 27–34%). The Wilcoxon rank-sum comparison against the unchanged post-implementation cohort remained highly significant ($p < 0.0001$), consistent with the primary analysis. In the restricted segmented regression model, the level change at the point of intervention remained large and highly significant (51.9 percentage points, 95% CI 49.3–54.5 percentage points, $p < 0.0001$), only modestly attenuated from the primary estimate (54.0 percentage points) but consistent in magnitude and direction. The pre-implementation slope point estimate was similar ($\beta = 0.233$ percentage points/month) but no longer reached statistical significance (95% CI −0.223 to 0.689, $p = 0.316$), likely a consequence of the reduced degrees of freedom rather than a qualitatively different secular trend. The change in slope at the point of intervention remained non-significant ($\beta = -0.156$, 95% CI −0.627 to 0.314, $p = 0.515$), and the post-implementation slope was unchanged ($\beta = 0.077$ percentage points/month), since this cohort was unaffected by the exclusion. These results indicate that the primary study conclusion that an abrupt, sustained level change at implementation rather than a continuation of a pre-existing secular trend is robust to exclusion of the 2022 observations.

Neither ED encounter volume nor weighted PPH demonstrated a significant level change or slope change coincident with the implementation date. Mean monthly ED volume was 27,667 (SD 1,060) pre-implementation versus 27,124 (SD 1,541) post-implementation ($p = 0.342$). Segmented regression estimated a pre-implementation intercept of 28,331 (95% CI 26,503–30,159) with a non-significant monthly slope of −132.9 encounters (95% CI −458 to 192, $p = 0.366$), and a post-implementation intercept of 29,029 (95% CI 28,114–29,943) with a significant monthly slope of −136.0 encounters (95% CI −193 to −79, $p < 0.0001$) – a rate essentially unchanged from the pre-implementation period. The level change at the point of intervention was 697.2 encounters (95% CI −1,171.7 to 2,566.1, $p = 0.453$) and the change in slope was −3.1 encounters/month (95% CI −300.0 to 293.8, $p = 0.983$), neither of which reached statistical significance.

Weighted PPH showed a similar pattern. The pre-implementation intercept was 2.02 (95% CI 1.94–2.10) with a non-significant monthly slope of −0.012 (95% CI −0.026 to 0.002, $p = 0.082$), while the post-implementation intercept was 1.92 (95% CI 1.84–2.00) with a near-zero monthly slope of 0.001 (95% CI −0.004 to 0.006, $p = 0.783$). The level change at intervention was −0.097 (95% CI −0.247 to 0.053, $p = 0.197$) and the slope change was 0.0128 (95% CI −0.011 to 0.037, $p = 0.281$), neither reaching statistical significance.

5. Discussion

This multi-site, quasi-experimental ITS study demonstrates that implementation of a non-interruptive, note-embedded EHR CDS tool was associated with a rapid, large, and sustained increase in HEART score monthly documentation compliance rising from a median of 29% pre-implementation to 82% post-implementation across six ED sites over four years of observation. Segmented regression confirmed a large

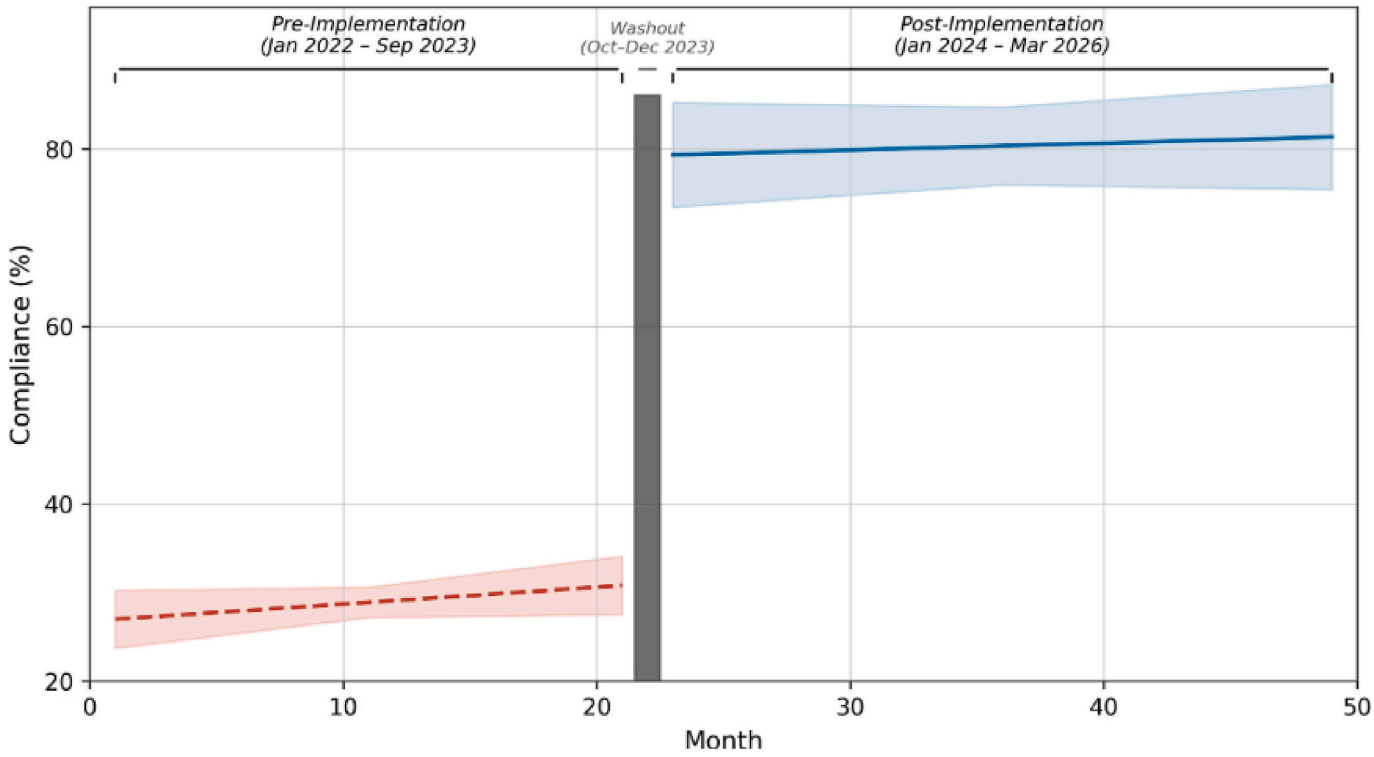


Fig. 2. Segmented regression fit for monthly HEART score documentation compliance (%) by cohort with 95% confidence limits.

abrupt level change at the point of intervention without a significant alteration in monthly trend slope, a pattern consistent with a durable step-change behavioral effect rather than a transient or attenuating response. This study provides the most longitudinally rigorous published evaluation of an EHR-embedded CDS tool targeting HEART score documentation to date and extends prior evidence for the effectiveness of non-interruptive CDS design in the ED setting.

The magnitude of the observed improvement is noteworthy. The 53 percentage-point absolute increase in median monthly compliance substantially exceeds the gains reported in most prior CDS implementation studies using process-level outcomes in the ED [9]. Furthermore, the plateau at 80–85% sustained over 27 months without appreciable monthly drift contrasts sharply with the attenuating improvements often observed following educational campaigns or interruptive alert-based interventions, where initial gains frequently erode due to alert fatigue and provider habituation [10,11]. The observation that pre-implementation compliance carried a modest but significant positive secular trend (approximately 0.2 percentage points/month) further supports the conclusion that the post-implementation gains represent a true intervention effect: even if this secular trend had continued unabated, it would have contributed only approximately 5 additional percentage points over 27 months, compared to the 53 percentage point observed level change.

The clinical significance of improved and sustained HEART score documentation is substantial. Systematic application of the HEART score has been associated with a 20–45% reduction in hospital admissions in prospective implementation studies, and the HEART Pathway achieves an NPV of 99.6% for 30-day MACE, enabling the safe early discharge of low-risk patients [15,16]. Consistent documentation as discrete EHR data elements creates the infrastructure for ongoing quality monitoring, cross-site benchmarking, and longitudinal outcomes research that was previously not feasible.

From an informatics perspective, these findings reinforce the CDS success principles identified by Kawamoto et al.: provision of support within existing clinician workflow, delivery at the point and time of decision-making, and actionable computer-based recommendations [21]. The note-embedded hyperlink design which required no additional navigation, no dismissal interaction, and no hard stop minimized workflow friction while delivering contextually appropriate guidance. Born et al.'s systematic review further supports the advantage of passive, analytically-timed information systems over active interruptive alerts in the time-pressured ED environment [20]. The high uptake rates achieved without interruptive alerts are consistent with findings from the EvidencePoint platform [22] and collectively reinforce that low-friction, workflow-embedded CDS design is achievable at multi-site scale using existing EHR infrastructure. Although results were aggregated across six ED sites with different patient volumes and case mixes, the consistent and durable improvement observed throughout the post-implementation period suggests the intervention effect is robust to site-level heterogeneity.

To further evaluate whether a concurrent change in ED operational volume or staffing-adjusted workload could account for the observed increase in documentation compliance, we performed a parallel segmented regression on monthly ED encounter volume and weighted PPH across the same pre- and post-implementation windows. Neither metric demonstrated a significant level or slope change coincident with the implementation date, arguing against ED volume or staffing as an alternative explanation for the observed effect.

5.1. Limitations

Despite the ITS design providing stronger causal inference than an uncontrolled pre-post study, this remains a single health system evaluation without an external concurrent comparator. The absence of a parallel control site precludes definitive attribution of all observed changes to the CDS intervention, although the stable pre-implementation trend and abrupt level change at the known implementation date are consistent with a causal effect. The manuscript does not formally inventory all concurrent quality improvement or educational initiatives that may have occurred across the health system during the implementation period. No co-interventions targeting HEART score utilization were identified by the study team; however, we cannot fully

exclude the possibility that other system-level changes contributed to the observed improvement. This study measured documentation compliance as a process outcome and did not evaluate downstream clinical endpoints, including changes in admission rates, length of stay, resource utilization, or accuracy of calculated HEART scores. Whether improved documentation translates to changes in patient outcomes requires further study. Increased documentation compliance is a necessary but insufficient marker of improved clinical decision-making; future work linking HEART score documentation to downstream care processes such as admission rates and early discharge of low-risk patients would be important to establish clinical impact. Additionally, while site-level aggregate data were extracted, a formal analysis of intervention effect consistency across individual ED sites was not performed; site-level variation in adoption rates may exist and should be examined in future evaluations. The study was conducted within a single health system using Epic as the EHR platform; generalizability to other EHR systems, organizational structures, or provider populations with different baseline behaviors may be limited. The core design principles of this intervention – embedding a non-interruptive prompt within an existing provider documentation template and linking it to a structured flowsheet – are conceptually transferable to other EHR platforms that support similar note-embedded hyperlink and flowsheet functionality; however, the technical implementation pathway and associated informatics effort would differ across systems and would require local validation. Individual provider-level data were not available, precluding analysis of variability in tool uptake across provider types, training levels, or shift patterns. The GEE ITS model assumed independence among monthly observations with robust standard errors; alternative autocorrelation structures were not formally compared. While the 27-month post-implementation window is the longest longitudinal assessment of such an intervention published to date, evaluation beyond this period is warranted to assess sustainability in even longer terms. The segmented regression model deliberately included only two covariates – a cohort (pre- versus post-implementation) indicator and observation number (time) – consistent with standard ITS and segmented-regression methodology, in which the pre-intervention trend itself serves as the counterfactual control for the intervention effect. This approach is well suited to slowly-evolving, facility-level confounders such as case mix or staffing levels. Such factors tend to drift gradually rather than shift abruptly at a single time point and would be expected to manifest as a change in the pre-implementation slope rather than as a large, discrete level change precisely coincident with tool deployment which is the pattern we observed. In a post hoc analysis using the same segmented-regression framework, system-aggregate ED encounter volume and a staffing-adjusted patients-per-hour metric showed no significant level change or slope change at the implementation date, which is reassuring against – though, given the system-aggregate rather than encounter-level nature of these data, cannot definitively exclude – confounding by ED volume or staffing changes. We also note that the inclusion criteria (specific ICD-10 codes for undifferentiated chest pain with mandatory EKG and troponin testing) already restrict the analytic population to a clinically homogeneous cohort, which limits, without eliminating, the degree to which system-wide case-mix variation could confound this process-level outcome.

6. Conclusion

Integration of a non-interruptive, workflow-embedded EHR CDS tool was associated with a rapid, large, and durable increase in HEART score documentation compliance across a multi-site ED system, rising from a median of 29% to 82%. Interrupted time series analysis confirmed an abrupt level change at intervention without a significant alteration in monthly trend slope, consistent with a durable, step-change behavioral effect persisting over 27 months of post-implementation observation. These findings demonstrate that non-interruptive CDS tools designed within the natural flow of clinical documentation can achieve sustained, high provider uptake while minimizing workflow disruption and alert fatigue in high-acuity settings. Future work should evaluate the impact of such implementations on downstream clinical outcomes, resource utilization, disposition decisions, and inter-provider variability in HEART score component documentation.

Previous presentations or publications

Portions of this work were previously presented at the American College of Emergency Physicians (ACEP) Research Forum, Salt Lake City, Utah, September 2024.

Availability of data and material

Due to institutional privacy restrictions, the data supporting the findings of this study are not publicly available but may be available from the corresponding author upon reasonable request and institutional approval.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude (Anthropic) to assist with literature organization, preparation of the Related Works section, and generation of Fig. 1 from source data. All AI-generated content was reviewed, verified, and edited by the authors. The authors take full responsibility for the content of the published work.

CRediT authorship contribution statement

Eric Boccio: Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Randy Katz:** Writing – review & editing, Supervision, Project administration, Data curation, Conceptualization. **Michael Weiss:** Writing – review & editing, Supervision, Investigation, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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